Chapter 1

Why Supersymmetry?

At the end of Chapter 0 we recorded the degrees of freedom of the various free fields in a ledger, and remarked in passing that supersymmetry requires the bosonic and fermionic entries of that ledger to balance. But before we start building supersymmetry, there is a question worth settling first: why should we care about a symmetry like this at all?

The question is not as idle as it sounds. Not one of the particles we have seen in nature has been found to have a "superpartner". The partner of the electron, for example, would be a spin-zero particle with exactly the electron's charge and mass. If such a thing existed, it would have turned up in the laboratory long ago. So supersymmetry, if it exists at all, must be a *broken* symmetry, and the partners must all be much heavier than the particles we know.

Why, then, have so many physicists spent so much effort on it for decades? This chapter gives four reasons, followed by a short history. Let me say clearly up front: none of the four reasons is a "proof". They are more like four independent clues that happen to point in the same direction. Whether nature has actually adopted supersymmetry is a question only experiment can answer.

1.1 The Hierarchy Problem

There is one very special particle in the Standard Model: the Higgs boson. It is the only fundamental scalar in the Standard Model, with a mass of about 125 GeV. It is special because the mass of a scalar particle behaves very badly under quantum corrections.

First look at a well-behaved example. In quantum electrodynamics the electron mass receives loop corrections. If we cut off the momentum integrals in the loops at some very high scale Λ (say, the place where new physics takes over), the correction comes out as

$$\delta m_e \sim \frac{\alpha}{\pi}\, m_e \ln \frac{\Lambda}{m_e}. \tag{1.1.1}$$

This result has two nice properties: it is proportional to m_e itself, and it depends on Λ only logarithmically. The first property has a beautiful reason behind it. If $m_e = 0$, the Lagrangian acquires an extra chiral symmetry: the left- and right-handed Weyl spinors can each be rotated by their own phase (recall Section 0.5.5). The mass term is precisely what breaks this symmetry, so the correction to the mass must be proportional to the mass itself; when the mass is zero, so is the correction. Even if you push Λ all the way up to the Planck scale, 10^{19} GeV, $\ln(\Lambda/m_e)$ is only about fifty, and after multiplying by α/π it does no harm at all.

Scalars have no such protection. For a complex scalar field H, the mass term $|H|^2$ is invariant under any phase transformation and breaks no symmetry, so nothing stops quantum corrections from pushing it up enormously. The correction to the squared Higgs mass generally looks like

$$\delta m_H^2 = c_1 \Lambda^2 + c_2\, m^2 \ln \frac{\Lambda}{m} + c_3\, m^2 + \cdots, \tag{1.1.2}$$

where m stands for the masses of other particles in the theory and the c_i are dimensionless coefficients, usually carrying a loop factor like $1/16\pi^2$. (The logarithmic term has to be multiplied by a mass squared, or the dimensions won't match.) All the trouble comes from the first term: it is proportional to Λ^2.

Let's look at an example. Suppose the Higgs field couples to a Dirac fermion f through a Yukawa coupling $-\lambda_f H\bar{f}f$. The fermion loop gives[1]

$$\delta m_H^2 \supset -\frac{|\lambda_f|^2}{8\pi^2}\Lambda^2. \tag{1.1.3}$$

If Λ really is the Planck scale, this term exceeds m_H^2 by more than thirty orders of magnitude (see the example below). To end up with the observed Higgs mass, the "bare" mass parameter in the Lagrangian must cancel against this correction to thirty-odd decimal places. That is not a logical contradiction, but it looks extremely unnatural: why would two huge numbers that have nothing to do with each other happen to be almost exactly equal? This is called the **hierarchy problem**, or the **naturalness problem**.

Now suppose the theory also contains a complex scalar S coupled to the Higgs through $-\lambda_S|H|^2|S|^2$. The scalar loop gives

$$\delta m_H^2 \supset +\frac{\lambda_S}{16\pi^2}\Lambda^2. \tag{1.1.4}$$

Pay attention to the signs: the boson loop is positive, the fermion loop negative. This relative minus sign is no accident; it goes back to the Grassmann anticommutation of Section 0.6.1: every fermion loop comes with an extra factor of -1. So if every Dirac fermion is accompanied by exactly two complex scalars, and the couplings satisfy $\lambda_S = |\lambda_f|^2$, the Λ^2 terms cancel completely:

$$2 \times \frac{|\lambda_f|^2}{16\pi^2}\Lambda^2 - \frac{|\lambda_f|^2}{8\pi^2}\Lambda^2 = 0. \tag{1.1.5}$$

That is exactly what supersymmetry does. A Dirac fermion consists of two Weyl spinors (Section 0.6.1), each Weyl spinor comes with a complex scalar partner (this is the chiral multiplet mentioned at the end of Section 0.7.4), and supersymmetry further guarantees that the couplings satisfy $\lambda_S = |\lambda_f|^2$. So in supersymmetric theories $c_1 = 0$, and the cancellation holds at every order of perturbation theory, with no adjustment by hand.

What remains is the logarithmic term. If supersymmetry is broken, the partner mass m_S and the original fermion mass m_f are no longer equal, and the correction becomes roughly

$$\delta m_H^2 \sim \frac{|\lambda_f|^2}{16\pi^2}\left(m_S^2 - m_f^2\right)\ln\frac{\Lambda}{m_S}. \tag{1.1.6}$$

For this not to be much larger than m_H^2, the mass difference between partners and the original particles cannot be too big: roughly, it should be within the TeV range. That is why people once expected to see supersymmetric particles at the Large Hadron Collider (LHC). So far the LHC has not found them, and the lower limits on the masses of the partners of the gluon and the quarks (gluinos and squarks) have been pushed to about 2 TeV or even higher. This puts "natural" supersymmetry under considerable pressure. Still, the hierarchy problem itself has not gone away, and supersymmetry remains its most thoroughly studied solution.

[1] The coefficients here and below are taken from the review by S. P. Martin (see Appendix C); deriving them requires loop calculations in quantum field theory, which this book does not go into. What matters for us is only the coefficient and sign of the Λ^2 term.

Example 1.1 How fine is the tuning?

Take $\lambda_f = 1$ (the top quark's Yukawa coupling is about this size) and $\Lambda = 10^{19}$ GeV. Roughly what is the ratio of the correction (1.1.3) to m_H^2?

Solution $|\delta m_H^2| \approx \Lambda^2/8\pi^2 \approx 10^{38}/79 \text{ GeV}^2 \approx 1.3 \times 10^{36} \text{ GeV}^2$, while $m_H^2 \approx (125 \text{ GeV})^2 \approx 1.6 \times 10^4 \text{ GeV}^2$. The ratio is about 10^{32}. In other words, the bare mass parameter and the loop correction must cancel to the 32nd significant figure to leave behind the Higgs mass we observe. □

1.2 Gauge Coupling Unification

The Standard Model has three gauge interactions, corresponding to the gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times U(1)$, each with its own coupling constant. We write them as $\alpha_i = g_i^2/4\pi$ $(i = 1, 2, 3)$. In quantum field theory a "coupling constant" is not actually constant: its effective value depends on the energy scale μ at which you probe it. This is called the **running** of the coupling. At one loop, the running is surprisingly simple:

$$\frac{1}{\alpha_i(\mu)} = \frac{1}{\alpha_i(M_Z)} - \frac{b_i}{2\pi} \ln \frac{\mu}{M_Z}. \tag{1.2.1}$$

That is, $1/\alpha_i$ plotted against $\ln \mu$ is a straight line. The slope is set by the coefficient b_i, and b_i depends only on which particles are in the theory: what gauge charges they carry and what their spins are. At $M_Z \approx 91$ GeV, experiments give

$$\alpha_1^{-1} \approx 59.0, \qquad \alpha_2^{-1} \approx 29.6, \qquad \alpha_3^{-1} \approx 8.5. \tag{1.2.2}$$

(Here α_1 uses the normalization customary in grand unified theories, $\alpha_1 = \frac{5}{3}\alpha_Y$, where α_Y is the hypercharge coupling.) The Standard Model gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; in the Minimal Supersymmetric Standard Model (MSSM), all the partner particles take part in the running too, and the coefficients become $(\frac{33}{5}, 1, -3)$.

Figure 1.1 shows the two cases side by side. The three Standard Model lines cross pairwise, but at three different places; in the MSSM, the three lines very nearly meet at a single point near $\mu \approx 2 \times 10^{16}$ GeV, where $\alpha^{-1} \approx 24$ (Problem 1.2). If you believe that at high enough energies the three interactions are different facets of a single interaction (that is the idea of grand unification), then their couplings ought to become equal at some scale. The Standard Model can't manage this; with the superpartners added, it works. It is worth stressing that the MSSM was not designed for this purpose.

In fairness, two more remarks. First, the MSSM lines in the figure assume all the partner particles are near M_Z; if they are heavier, say in the TeV range, the meeting point shifts a bit, but the quality of the unification is still very good. Second, three lines meeting at a point proves nothing by itself; it is just a pretty "coincidence". Still, in the history of physics, coincidences of this kind have often deserved to be taken seriously.

1.3 Dark Matter

Cosmological observations, especially the Planck satellite's precise measurements of the cosmic microwave background, give an inventory of what the universe is made of: ordinary matter,

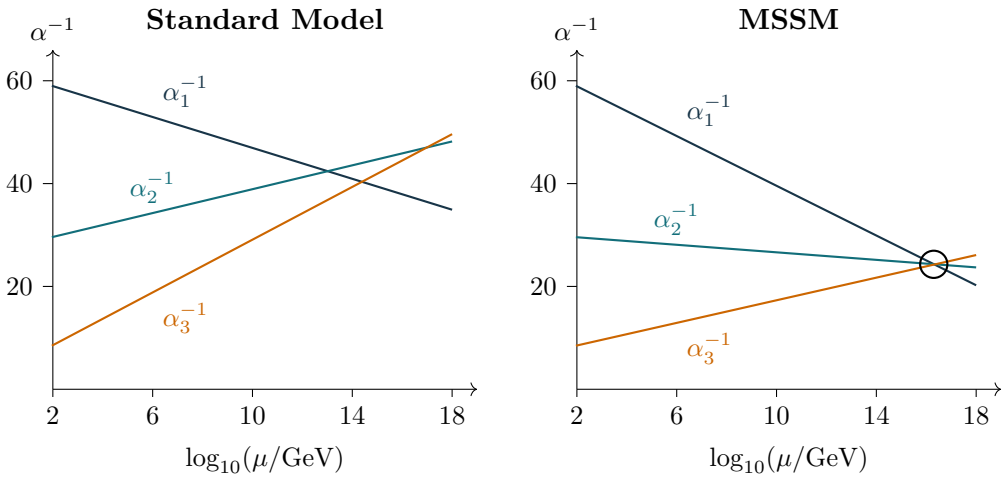


Figure 1.1: One-loop running of the inverse gauge couplings, computed from Eq. (1.2.1). Left: the Standard Model; the three lines cross pairwise, but at three different places. Right: the MSSM (for simplicity, all partner masses are taken to be near M_Z); the three lines very nearly meet at a single point (circled) at $\mu \approx 2 \times 10^{16}$ GeV.

meaning everything built from Standard Model particles, makes up only about 5%; dark matter about 27%; and the remaining 68% or so is dark energy. Dark matter has gravitational effects: galaxy rotation curves, gravitational lensing by galaxy clusters, and the fluctuations of the microwave background all require it. But it doesn't shine, and it barely interacts with ordinary matter. No particle in the Standard Model can play this role.

A good dark matter candidate should be stable (or at least much longer-lived than the age of the universe), electrically neutral, very weakly interacting, and slow-moving in the early universe ("cold"). Particles like this appear quite naturally in supersymmetric theories. To keep the proton from decaying too quickly, one usually requires a supersymmetric theory to preserve a discrete symmetry called **R-parity**: Standard Model particles have R-parity +1, and their superpartners −1. Then superpartners can only be produced and annihilated in pairs, and the lightest supersymmetric particle (LSP) has nothing to decay into, so it must be stable. In many models the LSP is a neutral fermion called the **neutralino**, a mixture of the partners of the photon, the Z boson and the Higgs. Better still, if a particle with a mass of a few hundred GeV to a TeV and interactions about as strong as the weak interaction was thermally produced in the early universe and then "froze out" as the universe expanded, the relic abundance you calculate comes out in the same ballpark as the observed dark matter density. This coincidence is often called the "WIMP miracle". (R-parity is a discrete $\mathbb{Z}_2$ symmetry; please don't confuse it with the continuous R-symmetry we will meet in Chapter 2.)

As with the LHC, underground direct-detection experiments have not yet seen a definite dark matter signal. So for now this clue remains just a clue.

1.4 A Laboratory Where Things Can Be Computed Exactly

The first three reasons all concern the universe we actually live in. The fourth is different: even if nature turns out not to use supersymmetry, supersymmetric theories are a superb laboratory for understanding quantum field theory.

Most of what we know how to compute in quantum field theory relies on perturbation theory: treat the coupling as small and expand order by order. But many of the most interesting phenomena happen where the coupling is strong, quark confinement for example: why does the strong interaction lock quarks inside hadrons forever? There perturbation theory fails completely. Supersymmetry constrains a theory so strongly that some physical quantities can be computed exactly, even at strong coupling.

Two examples. $\mathcal{N} = 1$ supersymmetric QCD (SQCD for short, the supersymmetric SU(N_c) gauge theory with quarks and their partners) was partially "solved" in the 1990s by Seiberg and others. We can't compute everything about it, but we can determine exactly which *phase* it is in for each number of quark flavors: confining, free, or conformal; it was even discovered that two theories that look completely different are in fact equivalent at low energies (Seiberg duality). This is something ordinary QCD still can't do.

At the other extreme is $\mathcal{N} = 4$ supersymmetric Yang–Mills theory (SYM), which has the largest amount of supersymmetry a four-dimensional gauge theory can have. Its β function vanishes identically; it is a conformal field theory. It would be premature to call it "completely solved", but many physical quantities can already be computed exactly for any value of the coupling. The prettiest example is the Wilson loop.

Imagine creating a static quark and antiquark a distance r apart at some instant, letting them coexist for a (Euclidean) time T, and then annihilating them. Their worldlines enclose a rectangular loop C of size $r \times T$. The Wilson loop $W(C)$ is the "phase factor" accumulated from the gauge field along this loop (we'll leave the precise definition for later). For large T, its expectation value behaves as

$$\langle W(C) \rangle \sim e^{-V(r)\,T}, \tag{1.4.1}$$

where $V(r)$ is the potential energy between the quark and the antiquark. So the Wilson loop tells us directly about the force between quarks: if, for instance, $V(r)$ grows linearly with r, the quarks are confined.

For $\mathcal{N} = 4$ SYM, the Wilson loop has been computed for a circular contour. In the limit where the number of colors $N \to \infty$ with the **'t Hooft coupling** $\lambda = g^2 N$ held fixed, its expectation value is exactly

$$\langle W_\circ \rangle = \frac{2}{\sqrt{\lambda}}\, I_1\big(\sqrt{\lambda}\big), \tag{1.4.2}$$

where I_1 is a modified Bessel function of the first kind. This formula was first obtained by summing a class of Feynman diagrams and conjecturing that all the other diagrams contribute nothing (Erickson, Semenoff and Zarembo; Drukker and Gross, 2000), and it was proved rigorously in 2007 by Pestun, using a technique (localization) that relies entirely on supersymmetry. Expanding it at weak coupling,

$$\langle W_\circ \rangle = 1 + \frac{\lambda}{8} + \frac{\lambda^2}{192} + \cdots, \qquad \lambda \ll 1, \tag{1.4.3}$$

each term corresponds to one order of Feynman diagrams (Problem 1.3). At strong coupling,

$$\langle W_\circ \rangle \approx \sqrt{\frac{2}{\pi}}\, \lambda^{-3/4}\, e^{\sqrt{\lambda}}, \qquad \lambda \gg 1. \tag{1.4.4}$$

That exponential factor $e^{\sqrt{\lambda}}$ is exactly what string theory in anti-de Sitter space gives in the AdS/CFT correspondence. One formula running all the way from perturbation theory to string theory: that is what "computable exactly" means.

Finally, a word in passing: supersymmetry is also an indispensable ingredient of string theory (the "super" in "superstring" comes from here); and if you make supersymmetry a local symmetry, the theory necessarily contains gravity; this is **supergravity**. These topics lie far beyond the present scope of this book, but they explain why supersymmetry is everywhere in theoretical physics.

***Problem 1.1** Suppose each Dirac fermion contributes Eq. (1.1.3) and each complex scalar contributes Eq. (1.1.4). (a) If each Dirac fermion is paired with N_S complex scalars, all coupling to the Higgs with strength λ_S, what is the condition for the Λ^2 terms to cancel? (b) Use the ledger of degrees of freedom in Section 0.7.4 to explain why supersymmetry gives exactly $N_S = 2$.

***Problem 1.2** Show from Eq. (1.2.1) that the lines for α_i^{-1} and α_j^{-1} cross where $\ln(\mu/M_Z) = 2\pi\big(\alpha_i^{-1} - \alpha_j^{-1}\big)/(b_i - b_j)$ (with both α^{-1} evaluated at M_Z). Using the numbers given in the text, find where α_1 and α_2 cross in the Standard Model, and where each of the three pairs of lines crosses in the MSSM.

Problem 1.3 Using the series $I_1(x) = \sum_{k=0}^{\infty} \frac{(x/2)^{2k+1}}{k!\,(k+1)!}$, verify the first three terms of the weak-coupling expansion of Eq. (1.4.2).

1.5 A Short History

1.5.1 Can Internal and Spacetime Symmetries Be Combined?

In the early 1960s, Gell-Mann and Ne'eman discovered that hadrons fit neatly into multiplets of a "flavor" symmetry group $\mathrm{SU}(3)_F$. For example, the eight spin-$\frac{1}{2}$ baryons (the proton, the neutron, the Λ, three Σ's and two Ξ's) form an octet, and the ten spin-$\frac{3}{2}$ baryons form a decuplet. A symmetry like this is called an **internal symmetry** (or a global symmetry): it acts on the "kind" of particle and leaves spacetime alone. Particles in the same multiplet have the same spin and similar masses.

A natural question followed: the spin-$\frac{1}{2}$ octet and the spin-$\frac{3}{2}$ decuplet look like members of one family; is there a larger symmetry that relates particles of different spin as well? In 1964, Gürsey, Radicati and Sakita combined $\mathrm{SU}(3)_F$ with the SU(2) that describes spin into SU(6), placing the octet and decuplet in a single 56-dimensional multiplet. The idea was quite successful; for instance, it predicted the ratio of the proton and neutron magnetic moments to be $-\frac{3}{2}$, compared with the experimental value of about -1.46. But the spin SU(2) is just the nonrelativistic rotation group. When people tried to extend it to a relativistic version, that is, to combine internal symmetry and the full Poincaré group nontrivially into one larger group, they ran into trouble at every turn. Was this a matter of doing it wrong, or was it simply impossible?

1.5.2 The Coleman–Mandula Theorem

In 1967 Coleman and Mandula gave the answer: it is impossible. Their theorem assumes:

1. The S-matrix is nontrivial: at almost all energies, two particles do scatter ($S \neq \mathbf{1}$);

2. Below any given mass there are only finitely many kinds of particle, and there is a mass gap (this rules out massless particles);
3. Scattering amplitudes are analytic functions of the energies and momenta;
4. The symmetry generators form a Lie algebra, that is, they obey only commutation relations among themselves.

Under these assumptions, the algebra of all symmetries can only be the direct sum of the Poincaré algebra and an internal symmetry algebra. That is, every generator T_a of an internal symmetry satisfies

$$\boxed{[P_\mu, T_a] = 0, \qquad [L_{\mu\nu}, T_a] = 0.} \tag{1.5.1}$$

T_a commutes with P_μ, so it doesn't change momentum, and hence doesn't change mass; T_a commutes with $L_{\mu\nu}$, so it is a Lorentz scalar and doesn't change spin. The conclusion: particles in the same internal multiplet must have exactly the same mass and spin. A symmetry like SU(6), which puts different spins into one multiplet, cannot be an exact symmetry of an interacting relativistic theory.

Why should this be? Here is an intuitive picture (the real proof is much longer). Consider elastic scattering of two particles. In the center-of-mass frame, conservation of momentum and energy fixes only the magnitude of the outgoing momenta; the scattering angle is still free to vary. Suppose there were an additional conserved quantity carrying Lorentz indices, say $\sum p_\mu p_\nu$ summed over the particles; it would have to be conserved too. That condition would pin down the outgoing direction as well, allowing only a few isolated scattering angles. But the scattering amplitude is an analytic function of the angle, and an analytic function that vanishes everywhere on an interval vanishes identically. So the only way out is a trivial S-matrix, in which the particles don't scatter at all. Extra conserved quantities with Lorentz indices cannot coexist with nontrivial scattering, and the only conserved charges that survive are Lorentz scalars, that is, Eq. (1.5.1).

1.5.3 The Loophole: Fermionic Generators

The fourth assumption of the theorem looks like a matter of course: don't symmetry generators always satisfy commutation relations? Actually, no. If some generators satisfy *anti*commutation relations among themselves, the so-called fermionic generators, the conclusion of the theorem no longer holds. Such a generator Q changes spin by $\frac{1}{2}$, turning bosons into fermions:

$$Q|\text{boson}\rangle = |\text{fermion}\rangle, \qquad Q|\text{fermion}\rangle = |\text{boson}\rangle. \tag{1.5.2}$$

Its multiplets contain particles of different spin, which is exactly what the Coleman–Mandula theorem forbids bosonic generators to do. The table below lists the key developments of the following years.

Year	Who	What
1971	Gol'fand, Likhtman	In Moscow, write down the first extension of the Poincaré algebra containing spinor generators Q_α. The short paper went almost unnoticed at the time.
1971	Ramond; Neveu, Schwarz; Gervais, Sakita	In string theory (then called "dual models"), independently find two-dimensional worldsheet Lagrangians with this extra symmetry.
1972	Volkov, Akulov	Obtain a nonlinear realization of supersymmetry from a different angle.
1974	Wess, Zumino	Write down the first four-dimensional supersymmetric quantum field theory: a complex scalar ϕ plus a Weyl spinor ψ_α, exactly the chiral multiplet of Section 0.7.4. From then on supersymmetry was widely known.
1975	Haag, Łopuszański, Sohnius	Generalize the Coleman–Mandula theorem to allow anticommutators, and prove that supersymmetry is the only possible extension.

The last row deserves a few more words. Haag, Łopuszański and Sohnius kept the first three assumptions of Coleman and Mandula and only allowed generators to have anticommutation relations among themselves. They proved: in four-dimensional spacetime, the fermionic generators can only be spin-$\frac{1}{2}$ spinors, $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$, with no spin $\frac{3}{2}$ or higher; there can be $\mathcal{N}$ copies of such spinors; for $\mathcal{N}>1$, so-called central charges can also appear; of the internal symmetries, only one kind (R-symmetry) can fail to commute with them; and if massless particles are allowed, the algebra can be enlarged further to a superconformal algebra. In a sentence: *under these very general assumptions, supersymmetry is the only possible nontrivial extension of spacetime symmetry.* That is the deepest reason for studying it. The next chapter looks concretely at what this unique extension looks like.

Problem 1.4 Suppose there is a bosonic conserved generator T that commutes with P_μ but not with $L_{\mu\nu}$. (a) Acting on a one-particle state $|p,s\rangle$, what would it preserve and what would it change? (b) What does the Coleman–Mandula theorem say about such a T? (c) The supersymmetry generators also commute with P_μ but not with $L_{\mu\nu}$ (Chapter 2 will prove this). Why don't they violate the theorem?
